

M09 / ASSESSMENT

M09 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

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CLOSED BOOK | SAVE ATTEMPT BEFORE ANSWERS

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Closed book. Save your answers before opening the protected answer guide. No forced timer.

1. Explain ETL vs ELT using one pipeline example and name the invariant both must preserve.
2. Put these in order and state each contract: raw, staging, intermediate, serving.
3. Calculate `net_sales` for `qty=2`, `unit_price=850`, `discount=0.10` and name two sources of nondeterminism to avoid.
4. The duplicate fixture has O6101 at 08:00 `qty1` and 09:00 `qty2`; O6102 has two conflicting rows at 08:30. What is accepted/discarded/quarantined and why?
5. A file renames `region` to `sales_region` and removes `discount_pct`. What should happen to raw, serving and state?
6. Reconcile the duplicate fixture with one equation. Then state two independent controls besides row count.
7. Why can a processed-file hash and a timestamp watermark both be useful but still insufficient for arbitrary late corrections?
8. Classify: database lock; 80% invalid data; failure after staging. For each: retry? target change? state advance?
9. Trace one serving order back to source evidence using run metadata.
10. Write the ten-step operating loop for the reliable batch pipeline in your own words.
11. In dbt, what is the difference between `source()` and `ref()`? Draw the supplied graph.
12. Name one generic and one singular dbt test. What does a passing data test mean?
13. Explain why `>= max(updated_at) plus unique_key='order_id'` helps at the current timestamp boundary but does not solve arbitrary late history.
14. Fresh design: a new source can resend corrected records up to 14 days late. Propose a safe incremental/reconciliation strategy and its cost.